



NUST CHIP DESIGN CENTRE

RISC-V Assembly

Weekly Task # 1
RISC-V Assembler

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1. In-Lab Tasks: (Write your lab task & screenshots here)

i. Test1:

Assembler output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> gcc -Wall -Wextra -std=c99 -O2 -o assembler assembler.c
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> ./assembler test.s output.bin -v
RISC-V RV32I Assembler
=====
Input file: test.s
Output file: output.bin

Pass 1: Building symbol table...
Symbol Table:
Name          Address
=====
main          0x00000000
EQ            0x00000018
LT            0x0000001c
LTU           0x00000020
NE            0x00000024
GE            0x00000028
GEU           0x0000002c

Pass 2: Generating machine code...
Machine Code (Hex):
Address      : Machine Code
=====
0x00000000: 0x00010c63
0x00000004: 0x00014c63
0x00000008: 0x00016c63
0x0000000c: 0x00011c63
0x00000010: 0x00015c63
0x00000014: 0x00017c63
0x00000018: 0x00000033
0x0000001c: 0x00000033
0x00000020: 0x00000033
0x00000024: 0x00000033
0x00000028: 0x00000033
0x0000002c: 0x00000033

Written 48 bytes to output.bin
Assembly completed successfully!
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> |
```

Venus output:

PC	Machine Code	Basic Code	Original Code
0x0	0x00010C63	beq x2 x0 24	beq sp,x0,EQ
0x4	0x00014C63	blt x2 x0 24	blt sp,x0,LT
0x8	0x00016C63	bltu x2 x0 24	bltu sp,x0,LTU
0xc	0x00011C63	bne x2 x0 24	bne sp,x0,NE
0x10	0x00015C63	bge x2 x0 24	bge sp,x0,GE
0x14	0x00017C63	bgeu x2 x0 24	bgeu sp,x0,GEU
0x18	0x00000033	add x0 x0 x0	add x0,x0,x0
0x1c	0x00000033	add x0 x0 x0	add x0,x0,x0
0x20	0x00000033	add x0 x0 x0	add x0,x0,x0
0x24	0x00000033	add x0 x0 x0	add x0,x0,x0
0x28	0x00000033	add x0 x0 x0	add x0,x0,x0
0x2c	0x00000033	add x0 x0 x0	add x0,x0,x0

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler\TestingApplication> python3 check.py hex_branches_only.dump output.hex
Files are identical
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler\TestingApplication> |
```

ii. Test2:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> ./assembler test.s output.bin -v
RISC-V RV32I Assembler
=====
Input file: test.s
Output file: output.bin

Pass 1: Building symbol table...
Pass 2: Generating machine code...
Machine Code (Hex):
Address    : Machine Code
=====
0x00000000: 0x00030f93
0x00000004: 0x00037613
0x00000008: 0x00b36293
0x0000000c: 0x01834293
0x00000010: 0x011e9e13
0x00000014: 0x003dde13
0x00000018: 0x018cae13
0x0000001c: 0xf00ebe13
0x00000020: 0x41eede13
0x00000024: 0x00012003
0x00000028: 0xffc12003
0x0000002c: 0xffa11003
0x00000030: 0xff815003
0x00000034: 0xff710003
0x00000038: 0xff614003

Written 60 bytes to output.bin
Assembly completed successfully!
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler>
```

0x8	0x00B36293	ori x5 x6 11	ori x5,x6,11
0xc	0x01834293	xori x5 x6 24	xori x5,x6,24
0x10	0x011E9E13	slli x28 x29 17	slli t3,t4,17
0x14	0x003DDE13	srli x28 x27 3	srli t3,s11,3
0x18	0x018CAE13	slti x28 x25 24	slti t3,s9,24
0x1c	0xF00EBE13	sltiu x28 x29 -256	sltiu t3,t4,-256
0x20	0x41EEDE13	srai x28 x29 30	srai t3,t4,30
0x24	0x00012003	lw x0 0(x2)	lw x0,0(sp)
0x28	0xFFC12003	lw x0 -4(x2)	lw x0,-4(sp)
0x2c	0xFFA11003	lh x0 -6(x2)	lh x0,-6(sp)
0x30	0xFF815003	lhu x0 -8(x2)	lhu x0,-8(sp)
0x34	0xFF710003	lb x0 -9(x2)	lb x0,-9(sp)
0x38	0xFF614003	lbu x0 -10(x2)	lbu x0,-10(sp)

iii. Test3:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> ./assembler
RISC-V RV32I Assembler
=====
Input file: test.s
Output file: output.bin

Pass 1: Building symbol table...
Symbol Table:
Name          Address
=====
main          0x00000000
boot          0x0000000c
multiply      0x00000010
exit          0x00000014

Pass 2: Generating machine code...
Machine Code (Hex):
Address      : Machine Code
=====
0x00000000: 0x00c000ef
0x00000004: 0x00c000ef
0x00000008: 0x00c0006f
0x0000000c: 0x00008067
0x00000010: 0x00008067

Written 20 bytes to output.bin
Assembly completed successfully!
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> 
```

PC	Machine Code	Basic Code	Original Code
0x0	0x00C000EF	jal x1 12	jal ra,boot
0x4	0x00C000EF	jal x1 12	jal ra, multiply
0x8	0x00C0006F	jal x0 12	jal x0, exit
0xc	0x00008067	jalr x0 x1 0	jalr x0,ra,0
0x10	0x00008067	jalr x0 x1 0	jalr x0,ra,0

iv. Test4:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> ./ass
RISC-V RV32I Assembler
=====
Input file: test.s
Output file: output.bin

Pass 1: Building symbol table...
Pass 2: Generating machine code...
Machine Code (Hex):
Address    : Machine Code
=====
0x00000000: 0x00030fb3
0x00000004: 0x400302b3
0x00000008: 0x00037633
0x0000000c: 0x000362b3
0x00000010: 0x000342b3
0x00000014: 0x000e9e33
0x00000018: 0x000dde33
0x0000001c: 0x000cae33
0x00000020: 0x000ebe33
0x00000024: 0x400ede33

Written 40 bytes to output.bin
Assembly completed successfully!
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler>
```

PC	Machine Code	Basic Code	Original Code
0x0	0x00030FB3	add x31 x6 x0	add x31,x6,x0
0x4	0x400302B3	sub x5 x6 x0	sub x5,x6,x0
0x8	0x00037633	and x12 x6 x0	and a2,x6,x0
0xc	0x000362B3	or x5 x6 x0	or x5,x6,x0
0x10	0x000342B3	xor x5 x6 x0	xor x5,x6,x0
0x14	0x000E9E33	sll x28 x29 x0	sll t3,t4,x0
0x18	0x000DDE33	srl x28 x27 x0	srl t3,s11,x0
0x1c	0x000CAE33	slt x28 x25 x0	slt t3,s9,x0
0x20	0x000EBE33	sltu x28 x29 x0	sltu t3,t4,x0
0x24	0x400EDE33	sra x28 x29 x0	sra t3,t4,x0

v. Test5:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> ./ass
RISC-V RV32I Assembler
=====
Input file: test.s
Output file: output.bin

Pass 1: Building symbol table...
Pass 2: Generating machine code...
Machine Code (Hex):
Address    : Machine Code
=====
0x00000000: 0x00012023
0x00000004: 0xfe011d23
0x00000008: 0xfe010ba3

Written 12 bytes to output.bin
Assembly completed successfully!
PS F:\muddassir_ali_siddiqui\ncdc\riscv\weekly_tasks\assembler> |
```

PC	Machine Code	Basic Code	Original Code
0x0	0x00012023	sw x0 0(x2)	sw x0,0(sp)
0x4	0xFE011D23	sh x0 -6(x2)	sh x0,-6(sp)
0x8	0xFE010BA3	sb x0 -9(x2)	sb x0,-9(sp)